Sharan S C

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EDUCATION

Bachelor Of Engineering: The Oxford College of Engineering, Bangalore 9.25 CGPA | 2023 – 2027

ACHIEVEMENTS

- 1st Place — **OxyHack**, Oxford (2026) & 2nd Place — **HackSetu**, BNMIT (2026): Built **ForeignPay** (fintech cross-border payments) and **SAHAJ** (AI-driven traffic signal coordination using Manhattan Distance & Relative Vectoring)
- Solved **880+ problems** on LeetCode with a peak contest rating of **1832 (Global Top 5%)**, consistently competing in weekly and biweekly contests · LeetCode Profile ↗
- First-authored and published **"FBG Sensor Design and Analysis for Early Detection of Cancer"** — *2024 IEEE INDIACom (11th International Conference on Computing for Sustainable Global Development)* · IEEE Xplore Profile ↗

SKILLS

Languages & Frameworks: Java, Python, C, JavaScript, Spring Boot, FastAPI, React
Cloud & Devops: AWS(EC2, RDS, LAMBDA), Docker, Kubernetes, GCP Compute Engine
Databases & Messaging : MySQL, MongoDB, PostgreSQL , Apache Kafka

EXPERIENCE

Software Development Intern | INTELAIX NOV 2025 – JAN 2026

Built backend infrastructure to migrate client communication from WhatsApp to scalable web push notifications

- Engineered a Java Spring Boot + Apache Kafka **event-driven pipeline** to replace manual WhatsApp notifications with automated **web push** delivery, integrating FCM token lifecycle management (registration, validation, dispatch)
- Architected for **10,000+** device registrations with Kafka consumers triggering **real-time** push notifications on system events — achieving sub-100ms **async processing** over synchronous blocking calls

PROJECTS

FreightIQ - Freight Marketplace & Intelligent Driver Matching | Microservices | GitHub | LIVE ↗

Java, Spring Boot, Python, FastApi, XGBoost, React, Docker, PostgreSQL

- Architected a 4-service microservices platform (User, Shipment, ML, Frontend) for an **India-focused** freight marketplace, fully containerized with Docker and deployable via single docker-compose up command
- Built an **XGBoost ML recommendation engine** (FastAPI) trained on **69,718 records** achieving **ROC-AUC of 0.91**, ranking top-3 drivers per shipment across 14 features including bid-to-budget ratio, on-time rate, and cargo compatibility
- Designed **20+ REST APIs** across Spring Boot services with Feign Client inter-service communication, enforcing a strict shipment lifecycle state machine (OPEN → ASSIGNED → IN_TRANSIT → DELIVERED → CANCELLED)
- Deployed full stack on **AWS EC2** with custom DuckDNS domain and SSL — ML service scores all bidders in a **single batched predict_proba call** via Python asyncio parallel fetching across 3 microservices

Sentinel – Reimbursement Fraud Detection System | Microservices | GitHub | LIVE ↗

Java, Spring Boot, OCR(tesseract), PostgreSQL, REST APIs, Docker, Kubernetes (Minikube)

- Designed two Spring Boot microservices — **Reimbursement Request Service** and **Employee Policy Management Service** enforcing clean separation of claim processing and policy validation, deployed live on **AWS EC2 with DuckDNS domain and HTTPS/SSL**
- Implemented end-to-end reimbursement workflow processing **JSON + receipt image uploads**, integrating **Tesseract OCR and AI-extraction** to cross-validate receipt data against claim details across full request lifecycle states
- Built a **5-layer fraud detection engine** — rule-based checks, historical spend analysis, **perceptual image hashing (pHash)**, and text hash similarity — generating fraud scores across **4 severity tiers** (LOW / MEDIUM / HIGH / CONFIRMED) with per-employee policy-limit enforcement